



Course Syllabus:
ESC-330-M: Engineering Thermodynamics I
Spring 2026
Albert Nerken School of Engineering
The Cooper Union for the Advancement of Science and Art

Instructor: Dr. Kamau Wright

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Office: 41 CS, 720* (or via Teams)

Office Hours: Mondays 2 p.m. to 3 p.m.
Tuesdays 9 a.m. to 10 a.m.
Thursdays 9 a.m. to 10 a.m.
or by appointment

Class Hours: **Locations/ Days/times:** 41 CS, 506* Tuesdays 10 a.m. to 11:50 a.m.
41 CS, 427* Thursdays 2 p.m. to 2:50 p.m.

*Location if instructor is remote: via Teams (unless otherwise stated)

Course Description: Rigorous development of the basic principles of classical thermodynamics. Zeroth, first and second laws of thermo-dynamics and their applications to open and closed systems. Analysis of thermodynamic processes, properties of real substances and thermodynamic diagrams.

Credits: 3

Prerequisites: Ph 112 (Physics I: Mechanics)

Textbook(s): Yunus Cengel, Michael Boles, "Thermodynamics: An Engineering Approach," 9th Edition, McGraw-Hill, ISBN: 978-1-260-04866-7

Supplementary: Sidebotham, G. "An Inductive Approach to Engineering Thermodynamics," Springer. 2022. ISBN-13: 978-3030204297. A pdf version is available through the Cooper library for free. Use the link:
<https://cooperunion.idm.oclc.org/login?url=https://link.springer.com/book/10.1007/978-3-030-20430-3>. Hard or soft copies can be purchased there if desired.

Course Outline:

Wk	Dates	Ch	Topics	HW	Assessment
1	1/20 1/22	1	Introduction and Basic Concepts (systems, control volumes, properties, pressure, temperature, states, equilibrium, processes, the Zeroth law)	• 1-58 Pressure • 1-70 Pressure, Manometer • 1-105 Review Problem • 1-118 FE Problem	
2	1/27 1/29	2	Energy, Energy Transfer, and General Energy Analysis (work, heat, the First Law)	• 2-17 Forms of Energy • 2-30E Mechanical forms of work • 2-45E The 1st Law of Thermo.	Quiz (Thu) *Quiz weeks may be moved
3	2/3 2/5	2	Energy, Energy Transfer, and General Energy Analysis (energy conversion, heat transfer)	• 2-92 Mechanisms of Heat Transfer • 2-119 Review Problem • 2-129 FE Problem	
4	2/10 2/12	3	Properties of Pure Substances (phase-change, property diagrams, thermodynamic property tables)	• 3-26E Property Tables • 3-37 Property Tables • 3-43 Property Tables	Quiz (Thu)
5	2/17 (mod) 2/19	3	Properties of Pure Substances (property tables, equations of state, ideal gas)	• 3-60 Property Tables • 3-69 Ideal Gas • 3-75 Ideal Gas	Project
6	2/24 2/26	3	Properties of Pure Substances (equations of state)	• 3-95E Other Equations of State • 3-136 FE Problem	Exam#1 (Thu, Ch. 1, 2, 3)
7	3/3 3/5	4	Energy Analysis of Closed Systems (moving boundary work, polytropic processes, specific heats, energy)	• 4-23 Moving Boundary Work • 4-44 Closed System Analysis • 4-54 Specific Heats, Δu and Δh	
8	3/10 3/12	4	Energy Analysis of Closed Systems (specific heats, internal energy, enthalpy, energy balance)	• 4-73E Energy analysis: Ideal Gases • 4-138 Review Problem • 4-148 FE Problem	Quiz (Thu)
9	3/17 - 3/19	← SPRING RECESS →			n/a
10	3/24 3/26	5	Mass and Energy Analysis of Control Volumes (flow work, nozzles, diffusers, turbines, compressors, throttling valves, <u>pumps</u>)	• 5-28E Nozzles and Diffusers • 5-47 Compressors • 5-57 Turbines • 5-63 Throttling Valves	Project *Project dates may be moved
11	3/31 4/2	5	Mass and Energy Analysis of Control Volumes (mixing chambers, heat exchangers, pipes, ducts, unsteady-flow processes)	• 5-73 Mixing chamber • 5-78E Heat Exchanger, Condenser • 5-185 Review Problem (unsteady) • 5-199 FE Problem	
12	4/7 4/9	6	The Second Law of Thermodynamics (thermal reservoirs, heat engines)	• 6-17 Heat Engines, Efficiency • 6-57 Refrigerators (cycles)	Exam#2 (Thu, Ch. 4, 5)
13	4/14 4/16	6	The Second Law of Thermodynamics (thermal efficiency, refrigerators, heat pumps, ir-reversible processes, Carnot)	• 6-84E Carnot Heat Engines • 6-103 Carnot Refrigerator • 6-106 Heat Pump	
14	4/21 4/23	6	The Second Law of Thermodynamics (Carnot)	• 6-135 Review Problem • 6-159 FE Problem	Quiz (Thu)
15	4/28 4/30	7	Entropy (increase of entropy principle, isentropic processes, T-s diagrams)	• 7-24 Increase of Entropy Principle • 7-93 Entropy Changes, Ideal Gases • 7-107 Reversible Steady Flow	
16	5/5	7, 9	Entropy (T ds Relations, entropy change, isentropic efficiencies) Gas Power Cycles (air standard cycle)	• 7-117E Isentropic Efficiencies • 7-136 Entropy Balance • 7-175 Review Problem • 9-13 Actual and Ideal Cycles	Quiz (Thu)
17	5/12 5/14		Final Exam (Mon); Final Project (Thu)— Report + Presentation		Finals

ABET Student Outcomes:

The Engineering Commission (EAC) of ABET has identified required student outcomes that support the program educational objectives associated with accredited engineering programs. Attainment of these outcomes prepares graduates to enter the professional practice of engineering. The outcomes are as follows:

- [ABET 1] an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- [ABET 2] an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- [ABET 3] an ability to communicate effectively with a range of audiences.
- [ABET 4] an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
- [ABET 5] an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
- [ABET 6] an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
- [ABET 7] an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Course Outcomes:

When students have completed this course, they will be able to:

1. Recall and explain the units, symbols, properties, and terminology of thermodynamics including concepts such as properties, states, temperature, pressure, systems, control volumes, equilibrium, processes, work, heat, and cycles. [ABET 1]
2. Apply the first and second laws of thermodynamics to open and closed systems undergoing a range of thermodynamic processes. [ABET 1]
3. Apply mass balances and energy balances to engineering devices that convert energy from one form to another.
4. Comprehend, describe, and communicate thermodynamics problems and their solutions with words, schematic diagrams, and mathematical expressions. [ABET 1, 2, 7]
5. Create schematics that include labels, which are useful to thermodynamics analysis. [ABET 1, 7]
6. Analyze and determine the properties and states of real substances, such as steam and refrigerant 134-a, and of ideal gases from either tabular data or equations of state. [ABET 1]
7. Demonstrate understanding of reversible and irreversible processes and plot processes on P-v and T-s diagrams. [ABET 1]
8. Evaluate the efficiency of heat engines and compare with the Carnot heat engine efficiency. [ABET 1]
9. Determine the coefficient of performance of refrigerators and heat pumps and compare with refrigerators and heat pumps operating on the reversed Carnot cycle. [ABET 1]
10. Evaluate modern processes and technologies using thermodynamic analysis. [ABET 1]

Assessment:

Students' achievement of the course outcomes and their final grade will be determined by their performance on two midterms (and quizzes, if applicable), project assignments, and a final exam. Any teamwork activities facilitated in this course will consider instructor assessment of contributions based on instructor observations and peer assessment.

<u>Grading policy:</u>	1.	Quizzes – no makeup; (lowest or missing quiz grade dropped)	20 %
	3.	Midterm Exams (Weeks 6 and 12) - (Two @ 20% each)	40 %
	4.	Project Assignments/Special Assignments	15 %
	5.	Final Exam (Comprehensive)	25 %
	Total		100 %

(The instructor reserves the right to modify the above distribution as required.)

Homework—
Activities,
Questions, and
Problems

Additional Homework Questions and Problems beyond the Activities

Homework “Questions and Problems” are assigned from the textbook but not collected nor graded. They are to be completed weekly, to help prepare for exams (and quizzes, where applicable). Solutions to select problems will be provided.

Additional assignments

Additional activities, assignments, and/or pop quizzes may be included as part of the course at the discretion of the instructor.

Exams and
quizzes:

Exams (and quizzes, if applicable) are to be completed by an individual (they are not group work). Each exam (and quiz) will emphasize the material covered in class. However, you are responsible for all the material assigned, whether covered in class or not, and for any additional material that the instructor covers in class discussions or by other presenters. No credit will be given for missed exams (or quizzes) unless you have a valid excuse substantiated by an appropriate document or prior instructor approval. A grade of zero will be awarded for an unexcused absence during an examination (or quiz). For an excused examination absence (e.g., illness or emergency) the comprehensive final examination will be adjusted proportionally (e.g., if a student is ill and misses Exam I, the comprehensive final will be weighted 40% instead of 25%). The choice of whether to permit a make-up examination (or quiz) rests solely with the instructor.

Formatting
solutions:

To solve problems for class, homework, quizzes, and exams, number and label each of your steps as you do the following: [1] Write your “**objective**.” [2] Sketch a “**schematic**” diagram. [3] List and define “**given**” information, including values and/or symbols for known and unknown variables. [4] Briefly (in 1 – 3 sentences) explain your intended “**approach**,” including assumptions and principles you will use to solve the problem. [5] Solve the problem symbolically (with variables, equations, and concepts), numbering and describing each step of your solution. This will usually require several labeled steps (i.e., 5, 6, 7, 8, 9 ...). [6] Then, plug in numbers with units to solve numerically. This may also require several labeled steps. Although there are times when intermittent calculations are suitable, in general, save the plugging-in-numbers steps until after the steps that establish what is being done conceptually and symbolically. [7] Check your solution (units, magnitude, and direction). [8] Briefly discuss the value and limitations of the solution. If you don’t believe in your solution, explain why, or suggest what should be revisited. [Note] Although there are eight steps listed here, those listed as steps “[5]” and “[6]” may require multiple steps that you should list in your work as 5, 6, 7, 8, 9, 10 and/or more. Then, those listed here as steps “[7]” and “[8]” would then simply be labeled numerically using the next consecutive numbers that you require, e.g., 11, 12.

Video Policy:

An audio and/or video recording of class lectures absent the express consent of the professor is prohibited. Video and audio recordings of class lectures may be part of the classroom activity. They are for educational purposes and may be made available to all students presently enrolled in the course. Recordings may not be reproduced or shared with those not in the class, nor uploaded to online environments.

Classroom
Eating Policy:

To promote a safe, clean, and distraction-free learning environment, the consumption of food is prohibited in the classroom during scheduled class sessions. A covered water bottle is permitted. Other covered drinks (i.e., tea, coffee) are also allowed. Exceptions to this policy may be granted for documented medical or accessibility accommodations. Students are expected to dispose of all waste appropriately and to maintain shared spaces and equipment in a responsible manner. When instructional activities take place in a laboratory setting, all food and beverages—including covered drinks—are strictly prohibited. Failure to comply with this policy may result in the removal of the item or dismissal from the classroom or laboratory.

Accommodations
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Students with disabilities or who need special accommodations for this class are to notify the Director of Student Care and Support to formally establish accommodations, provide me with a copy of your accommodation letter, and meet with me as soon as possible to discuss your needs so arrangements can be made. I am happy to discuss options toward the goal of mitigating physical or academic barriers based on disability and helping to facilitate your full participation in the course. Note that The Cooper Union has limited resources and extra lead time is required for such arrangements to be feasible. To receive accommodations for an exam, you must notify me in writing at least two weeks before the accommodations are needed, and you must also be registered with the Director of Student Care and Support (and/or Dean of Students if needed). Students will not be afforded any special accommodations retroactively, i.e., for academic work completed prior to disclosure of the disability to me. Support services for students are described [here](#).

Counseling
Services:

Counseling Services at The Cooper Union are coordinated through the Office of Student Affairs. Cooper's [counseling and mental health services website can be found here](#).

Title IX policy:

The Cooper Union Title IX policy on sexual misconduct can be found [here](#).

AI policy:

The use of generative AI or machine learning tools (such as ChatGPT or Dall-E 2 or Midjourney) in completing course assignments in this course is not required nor encouraged but will be allowed with appropriate citations, documentation, and verification. Any work submitted without referencing its source will be considered under academic dishonesty policies and will be subject to disciplinary action. Here is a note from ASME on use of authorship and AI tools in manuscripts submitted for their publications: “ASME is a member of the [Committee on Publication Ethics \(COPE\)](#) and has implemented COPE recommendations for authorship and AI Tools... Authors who use AI tools in the writing of a manuscript, production of images or graphical elements of the paper, or in the collection and analysis of data, must be transparent in disclosing in the Materials and Methods (or similar section) of the paper how the AI tool was used and which tool was used. Authors are fully responsible for the content of their manuscript, even those parts produced by an AI tool, and are thus liable for any breach of publication ethics. Please visit the COPE website for more information about [artificial intelligence and authorship](#).” Here is a note from Open AI (the developers of ChatGPT) on the need to verify their information: “While ChatGPT can be a valuable resource, it is essential to cross-verify any critical information or claims with reputable scholarly sources. AI-generated content may not always be accurate or up-to-date.” If you use AI for this course in completing any assignment, an Appendix must be submitted that includes citation (provide a reference), verification (how did you check the information?), and documentation (what were the prompts and responses?) for each entry. Yes, this may be long. AI citation formatting and details to include in an Appendix can be provided by the instructor.

Core Values:

The Core Values of the Albert Nerken School of Engineering apply to multiple aspects of this course. Our core values are the principles that guide our internal conduct and our relationship with the external world. Living these core values creates a culture within the Albert Nerken School of Engineering that supports students, faculty, and staff in all we do. Please join me in striving to live our Core Values:

RESPECT

We treat others as they would have us treat them. We support the vigorous and open debate of ideas within a community marked by mutual respect.

INTEGRITY

We value ethical behavior, integrity, and transparency in all aspects of life. We respect others; conduct ourselves ethically, honestly, and openly; honor our commitments; and fairly resolve ethical issues.

DIVERSITY & INCLUSION

We are committed to Peter Cooper's radical commitment to diversity and value, living and working in a diverse community. We value, encourage, and promote all aspects of human differences, fostering a culture that embraces a broad variety of personal circumstances, experiences, perspectives, and opinions.

INTERDISCIPLINARITY

We value interdisciplinary approaches and the intersection of art, architecture, humanities, and engineering. We collaborate across disciplinary boundaries to create innovative solutions to societal challenges.

TEAMWORK

We embrace a developmental culture, one in which cooperation and collaboration are cornerstones. Our faculty, staff, and students are collaborative, supportive, and serve as role models and mentors for professional and personal growth.

FREE

We are committed to Peter Cooper's vision of a free center of learning.

SOCIAL IMPACT

Through our engagement we instill a sense of social justice that translates into action. We inspire members of our community to apply their expertise and leadership for the benefit of society and humanity.

PARTNERSHIP

We are an engaged partner with industry, academia, government, and the communities in which we live. We are an indispensable partner with New York City.